

Homework — 2.1.4 Thinking Logically — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [3] — 1 mark per gap:

- (a) `score >= 70` — must use `>=`; "70 or above" includes exactly 70, so `>` earns nothing.
- (b) `score >= 40`
- (c) `else`

The order matters: 70+ is tested first, so the `elseif` is only reached when the score is below 70.

2. [2] — Two correct rows per mark:

Requirement	Boolean condition
At least 1.4 m	<code>height >= 1.4</code> (accept <code>height >= 140</code> if cm stated)
More than 500	<code>score > 500</code>
No more than the balance	<code>amount <= balance</code>
Falls below 18	<code>temp < 18</code>

The operator must match the wording exactly ("at least" → `>=`, "more than" → strict `>`, "no more than" → `<=`, "falls below" → strict `<`).

3. [4] — 1 mark per row; for a false row the mark needs **False** plus a valid correction.

Statement	True / False	Correction
<code>spent > 100</code> awards the discount at exactly £100	False	100 is not greater than 100, so exactly £100 does not qualify; the condition should be <code>spent >= 100</code> .
OR allows a withdrawal only when both limits are satisfied	False	OR is true when either limit alone is satisfied; the condition must use AND : <code>amount <= balance AND amount <= dailyLimit</code> .
<code>elseif</code> tested only when the first condition is false	True	—
Pump left unchanged at exactly 50%	True	— (neither <code>level < 20</code> nor <code>level > 80</code> is true, so neither branch runs)

4. [3] — All five rows correct = 3 marks; three or four rows = 2; one or two rows = 1.

i	levels[i]	levels[i] < 20 ?	levels[i] > 80 ?	pump	Output printed
0	15	True	not tested	ON	ON
1	50	False	False	ON	ON
2	85	False	True	OFF	OFF
3	70	False	False	OFF	OFF
4	10	True	not tested	ON	ON

Accept "—" or blank in the `> 80` column when the first condition is true (the `elseif` is not evaluated).
Key points: pump stays ON at level 50 and stays OFF at level 70 (no branch taken), so the outputs are ON, ON, OFF, OFF, ON.

5. [2] — 1 mark condition: `amount <= balance AND amount <= dailyLimit` (accept equivalent variable names). 1 mark FALSE outcome: the withdrawal is **refused/declined** and no cash is dispensed. The condition must use the **AND** operator.

Part B (6 marks)

6. [2] — 1 mark: the tax step uses the **gross pay** value produced by the previous step as its input. 1 mark: because of this data dependency it cannot run until gross pay has been calculated — the order of the two steps is fixed.

7. [2] — 1 mark: a **primary key** is an attribute (or set of attributes) that **uniquely identifies** each record in a table. 1 mark: a **foreign key** is an attribute in one table that refers to the primary key of another table, linking the two tables.

8. [2] — `00101101`. 1 mark for the correct binary value (101101), 1 mark for the correct 8-bit form with leading zeros. ($45 = 32 + 8 + 4 + 1$.)

Total: 14 + 6 = 20 marks